Reconstruction of Optical Image Time Series With Unequal Lengths SAR Based on Improved Sequence–Sequence Model

Tao Peng, Meiling Liu, Xiangnan Liu, Qian Zhang, Ling Wu, and Xinyu Zou

***Abstract*— Optical remote sensing time series are optimal for understanding and monitoring biochemical changes of key phenological parameters, which is essential for the assessment of vegetation health. However, due to cloud contamination, optical images often lack several days' to months' worth of information. Therefore, reconstructing optical time series based on the synthetic aperture radar (SAR) is necessary, which has the advantage of the production of continuous images under all weather conditions. In this study, an improved sequence-to-sequence (Seq2Seq) model was proposed, which integrated teacher forcing (TF) with the attention mechanism to handle input and output time series with unequal lengths. The proposed model could be used to reconstruct optical full-band time series using SAR time series data and reduce sample requirements based on the modification of the loss calculation method. To explore the spatiotemporal scalability of the proposed model, the test samples were divided into three categories: same time but different spatial (STDS) domain, different time but same spatial (DTSS) domain, and both different temporal and spatial (DTDS) domains. The major findings can be summarized as follows: 1) 72.7% of the generated Landsat 8 sequence values had an absolute error (AE) of less than 0.05; 2) the mean AE (MAE) of all bands was less than 0.0812; 3) the mean squared errors (MSEs) of all samples were lower than 0.015 regardless of the type of the test sample; and 4) TF was introduced to the model to improve the accuracy of the generated Landsat 8 sequence, yielding an increase of more than 5.97%. We concluded that the proposed model had good robustness both temporally and spatially. It performed well in the reconstruction of optical image time series, provided a basis for the use of time series pairs with unequal lengths, and can be applied to cloud removal and vegetation index reconstruction.**

***Index Terms*— Deep learning, sequence-to-sequence (Seq2Seq), spatiotemporal scalability, synthetic aperture radar (SAR), time series reconstruction, unequal.**

I. Introduction

HIGH time resolution and real-time data are required for the dynamic monitoring of surface vegetation parameters [1]. Understanding and monitoring biochemical changes of key phenological parameters throughout the life cycle of surface vegetation to determine the health of vegetation are essential for ecosystem stability and service [2]. In the current Earth observation system, time series analysis of remote sensing images is a low-cost method, which involves the analysis of a dataset with a high temporal and spatial resolution to understand and dynamically monitor physical phenomena at various growth stages [3], [4]. Furthermore, optical remote sensing data are advantageous because of their sensitivity in detecting biochemical components of vegetation and thus can be widely applied in vegetation monitoring [5], [6]. A variety of vegetation indices are based on optical remote sensing data [7], [8], [9]. However, optical remote sensing time series are incomplete because they are frequently corrupted by various types of missing information such as dead pixels, clouds, and cloud shadows [10]. These problems led to the establishment of the field of optical time series reconstruction based on multisource remote sensing data [11], [12], [13], [14], [15]. Two methods can be used to increase the temporal resolution of time series: image–image reconstruction and time series–time series reconstruction.

Image–image reconstruction is used to recreate missing or low-quality, monotemporal remote sensing images, replace them with the reconstructed image of the corresponding time position, and then connect them to obtain complete remote sensing time series. Image–image reconstruction methodologies can be divided into four categories [16]: spatial-based methods, spectral-based methods, temporal-based methods, and hybrid methods.

1) Spatial-based methods only rely on the information contained in the data itself, using spatial correlation to achieve the reconstruction of the missing region information [17]. Commonly used methods include interpolation methods [18], propagated diffusion methods [19], variation-based methods [20], and exemplar-based methods [21]. However, such methods are not effective when the data to be reconstructed happen in larger missing areas.
2) Spectral-based methods use data from other spectral bands as a complementary information source to achieve the reconstruction of a specific spectral band by modeling the relationship between the spectral bands. However, this method is usually suitable for dead pixel

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The authors are with the School of Information Engineering, China University of Geoscience, Beijing 100083, China (e-mail: 2004190016@cugb.edu.cn; liuml@cugb.edu.cn; liuxn@cugb.edu.cn; 3004210004@email.cugb.edu.cn; wuling@cugb.edu.cn; 3004190001@cugb.edu.cn).

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problems [22], but not for thick cloud processing problems.

3) Temporal-based methods use temporal images of the same region at different times as complementary information sources to reconstruct the lost pixels by modeling the relationships between the multitemporal data [23]. The major temporal-based approaches include the temporal replacement methods [24], [25], the temporal filter methods [12], and the temporal learning model-based methods [26], [27].
4) Hybrid methods utilize different spatial, temporal, and spectral information to jointly achieve the reconstruction of the missing information. The main state-of-the-art methods are spatiotemporal fusion [28], [29], [30], sparse representation-based methods [31], [32], [33], and deep learning-based synthetic aperture radar (SAR)-to-optical translation methods [34], [35], [36], [37], [38], [39], [40], [41], [42]. Although missing optical images are generated, postprocessing is required to obtain a complete time series, which is redundant from the perspective of computational efficiency.

In time series–time series reconstruction, the existing time series are used to directly reconstruct the target time series, which simplifies the reconstruction process. Time series–time series reconstruction methodologies can be divided into four categories: local interpolation, global interpolation, Gaussian process regression, and deep learning-based regression methods.

1) Local interpolation methods are based on the time evolution of a time series using a sliding time window. Commonly used methods include cubic spline interpolation [43], the weighted average method [44], the adapted local regression filter [45], and the Savitzky–Golay filter [46]. However, the sliding window size, which is a predefined parameter related to the gap length, directly affects the accuracy of these methods.
2) Global interpolation methods do not work on a local scale but restore lost information by fitting the data to a predefined function, mainly including the Whittaker smoother [47], [48], asymmetric Gaussian function fitting [49], [50], and Fourier-based harmonic analysis [12], [51], [52], [53]. In global interpolation methods, it is generally assumed that the data follow a certain distribution shape. Therefore, these methods lack flexibility when applied to nonstationary data, which is their main disadvantage [54].
3) Gaussian process regression methods are supervised, which can be used to capture any dependency between multiple related datasets [55]. However, Gaussian process regression methods are very sensitive to the kernel choice. Therefore, in the case of heterogeneous data, choosing a specific kernel size can limit the generalization capability. Simultaneously, the time sequence information of the input data cannot be learned.
4) To the best of our knowledge, direct time series reconstruction based on the deep learning regression method has only been successfully completed in two studies. In the first study, the multi-convolutional neural network (CNN)-sequence–sequence (Seq2Seq) model was applied [56], which presents a strategy to learn the correlation between optical and SAR time series based on the sequence of contextual information by extending the conventional CNNs and recurrent neural networks (RNNs). In the second study, the sentinels regression for vegetation monitoring (SenRVM) framework was used [57], which is a multisensor approach to regress SAR time series toward the normalized difference vegetation index (NDVI). The SenRVM framework uses a deep RNN architecture that integrates SAR acquisitions and ancillary data. However, while current deep learning-based regression methods consider the reconstruction of the NDVI time series, the whole band is not reconstructed, and determining the reconstruction ability of the deep learning method with respect to each band is highly improbable.

Optical images contain rich spectral information but are limited by poor weather conditions and cloud contamination [12], [14], [58]. The lack of optical images may last weeks to months or occur on critical dates involving important vegetation changes (e.g., the growth or senescence period or even the harvest period) [57]. Conversely, SAR is effective under all weather conditions, but the spectral information is insufficient for the identification of certain vegetation changes [59]. Recently, time series reconstruction using SAR images as auxiliary data has become a mainstream method [31], [60], [61], [62]. The SAR has the advantage of being weather-independent, eliminating cloud contamination. It has significant advantages over optically assisted images and broad application prospects in the field of remote sensing [63]. In the same time range, different sensors obtain different numbers of images due to the difference in the return period. When multisource sensors (this article specifically refers to SAR and optical time series) are used to reconstruct optical time series, lengths of the two-time series are not equal. In previous studies, two sets of time series pairs with unequal lengths were preprocessed to obtain time series pairs with equal lengths. However, this preprocessing step is time-consuming, laborious, and unnecessary. Therefore, establishing a method that can be used to process time series pairs with unequal lengths is necessary.

In this study, the Seq2Seq model was improved by integrating teacher forcing (TF) and the attention mechanism to address the aforementioned problems. The improved Seq2Seq model realized the reconstruction of optical time series data using SAR time series data and reduced the sample requirements based on the modification of the loss calculation method. The main contributions of this study are given as follows.

1) The improved Seq2Seq model can achieve cloud-free reconstruction from SAR time series to optical time series with unequal lengths.
2) The application of TF to the model improves the accuracy of the reconstruction.
3) The improved model has a certain anti-interference ability due to the lack of abnormal training data and has good spatiotemporal scalability.

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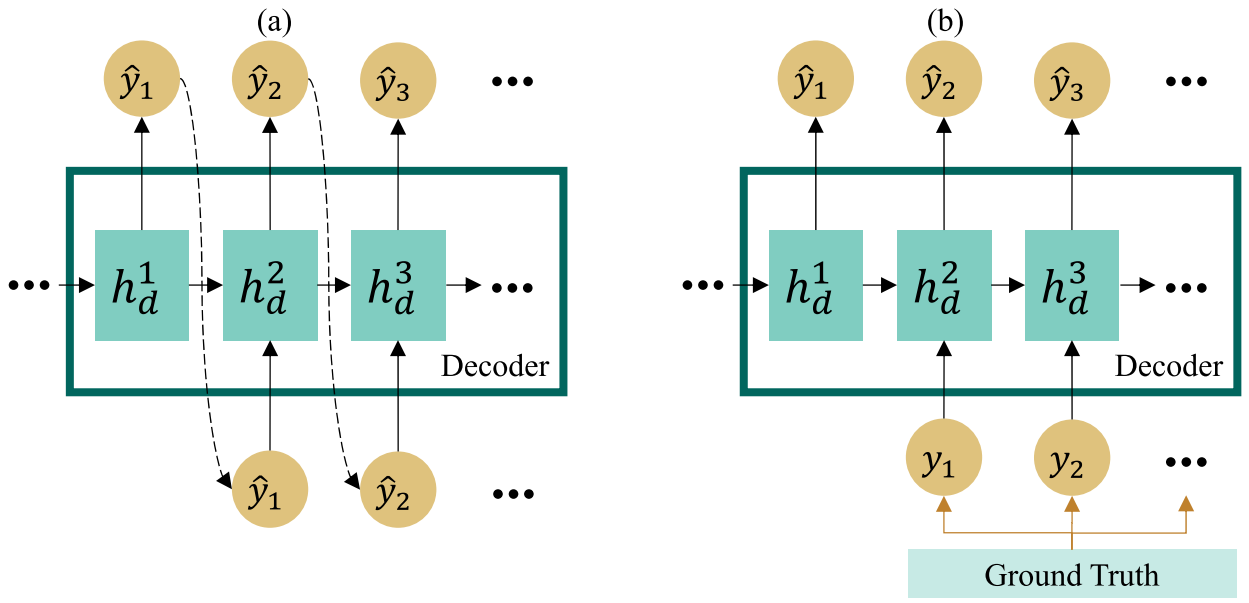


Fig. 1. Graphical illustration of two training strategies. y: real value. $\hat{y}$: generated value. h_d: hidden unit state of the decoder. (a) Without TF. (b) With TF.

The remainder of this article is organized as follows. In Section II, we review the relevant fundamentals and elaborate on the proposed methodology. Section III gives a brief description of the study area and data sets and details the implementation of the experiments. Experimental results are presented in Section IV. Section V discusses the main findings and the problems to be investigated further. Finally, the conclusion and prospects are presented in Section VI.

II. Materials and Methods

A. Fundamentals

1) Gated Recurrent Unit: Gated recurrent unit (GRU) adaptively captures the dependencies of different time scales to address long-term memory and gradient problems with respect to backpropagation and was proposed by Cho et al. [64]. Compared to the long short-term memory (LSTM) [65], the GRU does not contain separate memory cells [66] but implements a strategy that allows for both forgetting and selective memory in a single gating unit, thus offering the advantage of fewer parameters and lower computational complexity along with high performance.

2) Sequence-Sequence: The Seq2Seq [67] model essentially combines two RNNs with the help of encoder and decoder structures to achieve unequal lengths of the input and output sequences. In the encoder phase, the input sequence is read and represented as a context vector. The decoder phase estimates the generation probability of the output sequence with the context vector as the input [68].

3) Attention Mechanism: The attention mechanism [69], [70] has been illuminated in the fields of natural language processing and computer vision. The attention mechanism is a special structure embedded in machine learning models that automatically learns and calculates the contribution of input data to output data. This empowers the model to distinguish and discriminate, finding meaningful information from the clutter.

4) Teacher Forcing: TF [71] is a strategy that can be used for one-step-ahead prediction. It was originally developed for the training of an RNN as an alternative technique to backpropagation, which is widely used in autoregression. All tokens of the input are sent to the RNN and the loss is calculated and later backpropagated with a relative partial loss. However, this approach is problematic—if the outputs at the beginning of the training are inaccurate, the chances of the following outputs being correct decrease because they depend on previous erroneous states [see Fig. 1(a)]. The TF either uses the outputs of the previous states or the ground truth as input at each token in the training process [see Fig. 1(b)].

B. Proposed Model

To determine the correlation between Sentinel-1 and Landsat 8 sequences, an improved Seq2Seq model [see Fig. 2] was proposed in this study, which integrated TF and the attention mechanism and thus could be applied to remote sensing sequence reconstruction. As mentioned in Section I, trivial preprocessing steps are required in order to obtain sample pairs with equal-length time series from different numbers of images acquired by different sensors in the same time range. The Seq2Seq model proposed in this study serves as a framework capable of handling time series of different lengths and can be used more flexibly in the field of time series reconstruction.

Overall, the improved Seq2Seq model [see Fig. 2] consists of an encoder that transforms the input SAR feature sequence into feature vectors and a decoder that deciphers the feature vectors into output-generated Landsat 8 sequences of the corresponding periods. Then, the objective function of the model can be computed as

$$p\left(\hat{L}_1,\ldots,\hat{L}_t,\ldots,\hat{L}_m \mid S_1,\ldots,S_T,\ldots,S_n\right) = \prod_{t=1}^{m} p\left(\hat{L}_t \mid O,\hat{L}_1,\ldots,\hat{L}_t,\ldots,\hat{L}_m\right). \quad (1)$$

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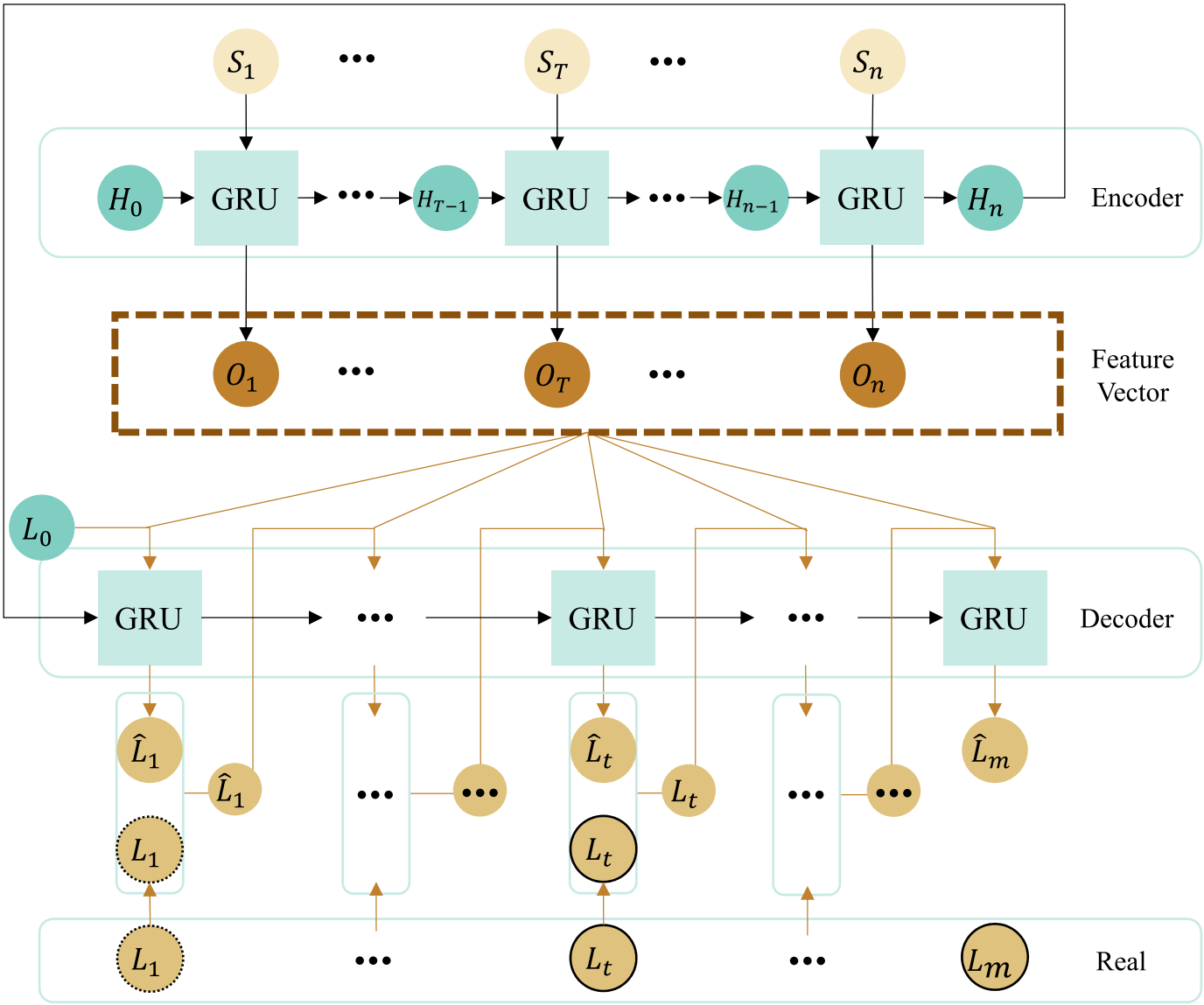


Fig. 2. Overall network architecture of the proposed model.

In this equation, the conditional probability $p(\hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m \mid S_1, \ldots, S_T, \ldots, S_n)$, where $S_1, \ldots, S_T, \ldots, S_n$ is the input SAR feature sequence and $\hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m$ is its corresponding output-generated Landsat 8 sequences whose length m may differ from n. Each $p(\hat{L}_t \mid O, \hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m)$ distribution is represented with a softmax over all the inputs in the SAR feature sequence, and $O = [O_1, \ldots, O_T, \ldots, O_n]$ is encoder output feature vectors at all moments. T represents the Tth moment $(1 < T < n)$ in the input SAR feature sequence and the encoder output feature vectors, and t represents the tth moment $(1 < t < m)$ in the output-generated Landsat 8 sequences and real Landsat 8 sequences.

1) GRU-Based Encoder: In the encoder phase, the information from the entire Sentinel-1 feature sequence was extracted for the subsequent attention-based decoder and then converted into an information vector. In contrast to discrete and non-specific word numbering, Sentinel-1 features were continuous and had specific physical meanings. Therefore, the embedding layer commonly used in natural language processing was discarded in this study. The encoder part was an iterated GRU with a hidden size of 2048.

At each moment T, the input of the GRU was the Sentinel-1 feature of the present moment S_T and a hidden unit state H_{T-1} passed down from the previous moment, which contains the relevant information of the previous node. Also, the output of the GRU was the hidden unit state at the current time H_T and the output feature vector O_T at the corresponding time.

If T is not the last moment of the SAR feature sequence, the output hidden unit state H_T will be passed to the encoder at the next moment, while if it is the last moment, the hidden unit state H_T will be passed to the decoder as the context vector. It is also noticeable that if T is the first initial moment, there will be no hidden state H_{T-1} from the previous moment. In this case, the general solution is to pass a randomly generated array or an all-zero array into the encoder as the initial hidden state H_0. In this study, a zero array is used as the initial input.

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In the internal structure of GRU, there is a reset gate (r) and an update gate (z). When data are passed to them, there is a procedure of taking a linear sum between the existing state and the newly computed state with exposing the whole state each time [66]. The detailed calculation of the passing process of the GRU is given as follows.

First, the gating signal r_T of the reset gate is obtained by calculating the Sentinel-1 feature of the present moment S_T and the hidden unit state H_{T-1} of the previous moment using

$$r_T = \sigma(W_r S_T + U_r H_{T-1}) \tag{2}$$

where σ is the sigmoid function by which the data can be transformed to a value in the range 0–1 to act as a gating signal.

Subsequently, the candidate activation H' is obtained by computing the gating signal r_T obtained in the previous step, S_T and H_{T-1} using

$$H^{'} = \tanh\left(W S_T + U\left(r_T \odot H_{T-1}\right)\right) \tag{3}$$

where tanh is an activation function to deflate the data to a range of -1 to 1 and $\odot$ is an elementwise multiplication. When OFF (r_T close to 0), the reset gate effectively makes the unit act as if it is reading the first symbol of an input sequence, allowing it to forget the previously computed state.

Similar to the first step, the gating signal z_T of the update gate is obtained by calculating S_T and H_{T-1} using

$$z_T = \sigma(W_z S_T + U_z H_{T-1}). \tag{4}$$

Finally, the activation H_T of the GRU at time T is a linear interpolation between the previous activation H_{T-1} and the candidate activation H'

$$H_T = (1 - z_T) H_{T-1} + z_T H^{'}. \tag{5}$$

By repeating the above four steps until all moments of the SAR feature sequence have been processed, the encoder obtains the final output feature vector $O = [O_1, \ldots, O_T, \ldots, O_n]$ and the context vector H_n. The output of the encoder is then passed to the subsequent decoder for Landsat 8 sequence generation.

2) Attention-Based Decoder: In the original decoder, the context vector H_n is used as an input to the decoder, which is capable of converting the context vector H_n into the output-generated Landsat 8 sequences $\hat{L} = \hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m$.

However, passing only the context vector H_n between the encoder and the decoder would make H_n responsible for the entire process of encoding the output-generated Landsat 8 sequences, as well as recording information on the moment-to-moment differences between the SAR sequences and the Landsat 8 sequences. Therefore, the attention mechanism was introduced to the decoder to reduce the requirements of the context vector H_n.

The attention mechanism forces the decoder to "pay attention" to the encoder's output feature vector $O = [O_1, \ldots, O_T, \ldots, O_n]$ when generating Landsat 8 sequence values for each moment. It calculates the weight according to the input and hidden state of each moment and then combines the output of the encoder at each moment with the input of the decoder at that moment. This means that the decoder will consider the Landsat 8-generated value and hidden state values at the previous time and combine each output corresponding to the part of Sentinel-1 of the encoder when generating the Landsat 8 value at a certain time. When combining the output of each moment, the attention mechanism will focus on the output of the encoder at one or more moments near the current time based on the moment of Landsat 8 that is currently needed to be generated. The introduction of the attention mechanism enabled the model to focus on multiple outputs of Sentinel-1 at different times when generating the Landsat 8 value, which can solve the problem of unequal intervals between the Sentinel-1 and Landsat 8 sequences.

The overall structure of the attention-based decoder is shown in Fig. 3. At each moment t, the inputs to the decoder are the hidden unit state h_{t-1} generated by the decoder at the previous moment, the Landsat 8 feature vector $\hat{L}_{t-1}$ generated by the decoder at the previous moment or the real Landsat 8 feature vector L_{t-1}, and the encoder's output feature vector $O = [O_1, \ldots, O_T, \ldots, O_n]$. Notably, when the attention-based decoder is at its initial moment, the context vector H_n generated by the encoder will be used as the initial hidden state.

The decoder is initiated with a full connection layer to convert the cloudless Landsat 8 real value L_{t-1} or output-generated Landsat 8 values $\hat{L}_{t-1}$ (depending on the probability of using TF) at the previous moment into feature $\tilde{L}_{t-1}$ (6). Subsequently, a dropout layer (7) with a probability of 0.5 was generated to avoid overfitting and enhance model generalization

$$\tilde{L}_{t-1} = \text{dropout}\left(W_{\text{input}} * L_{t-1} + b_{\text{input}}\right) \tag{6}$$

where dropout(x) is calculated as follows:

$$\text{dropout}(x) = x \odot b_m = \left(x_1 b_{m_1}, x_2 b_{m_2}, \ldots,\right) \tag{7}$$

where $\odot$ is the matrix multiplication and $b_m = (b_{m_1}, b_{m_2}, \ldots,)$ is the random mask vector with the following equation:

$$b_{m_i} = \begin{cases} 0, & r_i < p \\ \dfrac{1}{1-p}, & r_i \geq p \end{cases} \tag{8}$$

where r_i is a randomly generated decimal number between 0 and 1 and p is a predefined random deactivation probability, $p = 0.5$ in this study.

In the second step, the attention weight $\text{Attn}_{\text{weights}_{t-1}}$ was calculated: both the feature vector $\tilde{L}_t$ and the hidden unit state at the current moment h_{t-1} were input into Attn and Softmax layers

$$\text{Attn}_{\text{weights}_{t-1}} = \text{Softmax}\left(W_{\text{Attn}} * \left[\tilde{L}_{t-1}, h_{t-1}\right] + b_{\text{Attn}}\right) \tag{9}$$

where Softmax(x) is calculated as follows:

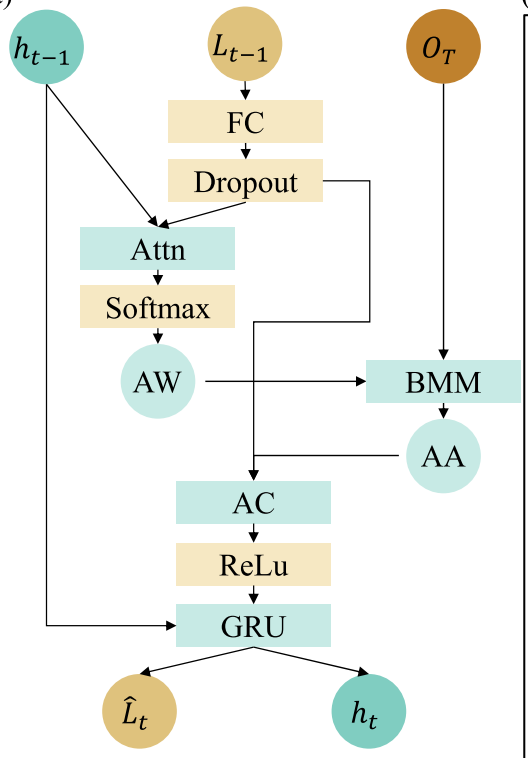
$$\text{Softmax}(x) = \frac{e^{x_1}}{\Sigma_{e=1}^{k} e^{x_i}}, \frac{e^{x_2}}{\Sigma_{e=1}^{k} e^{x_i}}, \ldots, \frac{e^{x_k}}{\Sigma_{e=1}^{k} e^{x_i}} \tag{10}$$

where x_i is the ith number in the x-vector and k is the length of the x-vector.

In the third step, the attention weight $\text{Attn}_{\text{weights}_{t-1}}$ and encoder output at all moments $[O_1, \ldots, O_T, \ldots, O_n]$ were

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(a)



FC: Full Connection Layer
BMM: Batch Matrix Multiplication
AW: *Attn_weights*
AA: *Attn_applies*
AC: *Attn_combine*

(b)

Input: $L_0, L_1, \ldots, L_t, \ldots, L_m; H_n; O_1, \ldots, O_T, \ldots, O_n$;
$Cloud_mask = \{c[0], c[1], \ldots, c[m]\}$
Initialize: n=number of Sentinel-1 time;
m= number of Landsat 8 time
do{
$if\ t = 0$:
$h_t = H_n$
$If\ c[t]! = False\ \&\ If\ use_{TF} = True$
$L_t = L_t$
$Else$:
$L_t = \hat{L}_t$
$\tilde{L}_t = dropout(W_{input} \cdot L_t)$
$Attn = W_{Attn} \cdot [H_t, \tilde{L}_t]$
$Attn_weights = softmax(Attn)$
$Attn_applies = Attn_weight \cdot [O_1, \ldots, O_T, \ldots O_n]$
$Attn_combine = W_{Attn_combine} \cdot [\tilde{L}_t, Attn_applies]$
$r_t = \sigma(W_r \cdot [h_t, Attn_combine])$
$z_t = \sigma(W_z \cdot [h_t, Attn_combine])$
$\tilde{h}_t = tanh(W_{\tilde{h}} \cdot [r_t * h_t, Attn_combine])$
$h_{t+1} = (1 - z_t) * h_t + z_t * \tilde{h}_t$
$\hat{L}_{t+1} = \sigma(W_o \cdot h_{t+1})$
$t = t + 1$
}while(t<m)
Output: $\hat{L}_0, \hat{L}_1, \ldots, \hat{L}_t \ldots, \hat{L}_m$

Fig. 3. Specific process and pseudocode representation of the decoder. (a) Attention-based decoder iteration process. (b) Pseudocode of attention-based decoder iteration process.

multiplied, and then, the product $\text{Attn}_{\text{applies}_{t-1}}$ was combined with feature $\tilde{L}_{t-1}$.

$$\text{Attn}_{\text{combine}\,t-1} = W_{\text{Attn_combine}} * \left[\text{Attn}_{\text{applies}_{t-1}}, \tilde{L}_{t-1}\right] + b_{\text{Attn_combine}}. \quad (11)$$

The combined results $\text{Attn}_{\text{combine}_{t-1}}$ are activated using the hyperbolic tangent (tanh) activation function with the following equation:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \quad (12)$$

Subsequently, the activated results and hidden unit state at the current moment h_{t-1} were input into the GRU to obtain the corresponding output $\hat{L}_t$ and hidden unit state at the current moment h_t. This process is similar to the update of the GRU in the encoder and will not be repeated here.

Finally, the output was converted into six bands from Landsat 8 based on a full connection layer. The final result of this decoder was the output-generated Landsat 8 sequence $\hat{L}$.

3) Loss Function: To supervise the proposed model, the network parameters are estimated in the training process by minimizing the loss function $\mathcal{L}$ (see Fig. 4). Given N training samples, $\mathcal{L}$ is defined as the square loss function (L2 LOSS) ℓ and submitted in the forward training propagation step as follows:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \ell\left(\hat{L}_i - L_i\right) \quad (13)$$

where ℓ is used to evaluate the average error between the output-generated Landsat 8 sequence values $\hat{L}_i = [\hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m]_1, \quad [\hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m]_2, \ldots, [\hat{L}_1, \ldots, \hat{L}_t, \ldots, \hat{L}_m]_N$ and real Landsat 8 sequence values $L_i = [L_1, \ldots, L_t, \ldots, \quad L_m]_1, [L_1, \ldots, L_t, \ldots, L_m]_2, \ldots, [L_1, \ldots, L_t, \ldots, L_m]_N$.

Considering the sensitivity of ℓ to outliers, the cloudy real Landsat 8 acquisitions were deleted from the l calculation. The equation includes cloud mask information $\text{CM} = [\text{CM}_1, \text{CM}_2, \ldots, \text{CM}_m]$. The validity flag of $\text{CM}_t \in (T, F)$ is associated with each time

$$\ell = \frac{1}{m}\sum_{t=1}^{m} \text{CM}_t |\hat{L}_t - L_t|^2. \quad (14)$$

4) TF Utilization Improvements: Using TF means that there is a certain probability that a real value will be used as the input for the current moment at every moment. This study pioneered the use of the TF, which was originally used only

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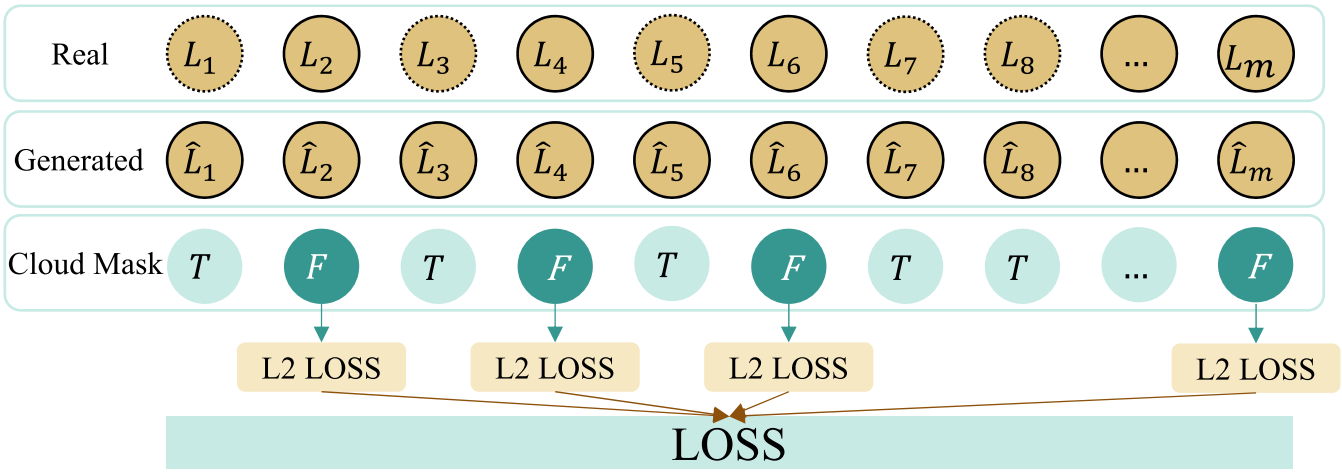


Fig. 4. Loss calculation method based on the cloud mask sequence.

for the model training process, for both the training and use of the model.

During model training, if real values at every moment are dependent on (using TF with 100% probability), it will lead to an overfitting problem, namely, the trained model will work well only for the training samples. However, if no real values are used at all (using TF with 0% probability), the model will hardly converge and may even converge in the wrong direction. Therefore, real values of a defined probability are used. During the training of the model in this study, this probability was set to 70%, thus reducing the accumulation of errors and speeding up the convergence of the model.

The TF method was also introduced in the process of using the model, but differently than in the model training. First, the overfitting problem does not need to be considered. Second, the introduction of real values will help the model generate more accurate feature values. Therefore, during the use of the model in this study, the probability of TF is set to 100%, so the generated values can be constrained to a certain extent to improve the accuracy of the results.

In optical time series reconstruction problems, we can obtain some cloud-free optical remote sensing images as real values. We conducted two sets of experiments to verify the feasibility of using TF in the model use phase: 1) the TF was used, i.e., whenever the cloudless Landsat 8 real-value sequence was available, the real value was used to guide the generation of the sequence at that moment, and 2) the TF was not used, i.e., the cloudless Landsat 8 real-value sequence was not used to guide the generation of the output-generated Landsat 8 sequence at that moment regardless of the existence of a cloudless Landsat 8 real-value sequence.

C. Evaluation Methods

To quantitatively evaluate the accuracy, five indicators were adopted in this study: absolute error (AE), mean AE (MAE), mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM). The AE is a measure of the degree of deviation between the generated and real values. The MAE is a measure of the error between pairs of observations of the same phenomenon, with a range of values from zero to positive infinity. The MSE measures the deviation between the generated and real values and detects the consistency between the two values. The smaller the MSE is, the denser the data, and the higher the quality of the generated sequence. The PSNR indicates image pixel statistical information, with a higher PSNR generally indicating higher image quality. The SSIM represents structural information by calculating the structural similarity between generated images and real optical images. Its value ranges from zero to one and will reach one when the two images are identical.

To simplify, we assume that we already have N samples of real value y and generated value $\hat{y}$. The equations for the calculation of the AE, MAE, MSE, PSNR, and SSIM are given as follows:

$$\mathrm{AE} = |y - \hat{y}| \tag{15}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=0}^{N} |y_i - \hat{y}_i| \tag{16}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=0}^{N} (y_i - \hat{y}_i)^2 \tag{17}$$

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right) \tag{18}$$

$$\mathrm{SSIM}(y, \hat{y}) = \frac{(2\mu_y\mu_{\hat{y}} + c_1)(2\sigma_{y,\hat{y}} + c_2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + c_1)(\sigma_y^2 + \sigma_{\hat{y}}^2 + c_2)} \tag{19}$$

where MAX_I is the maximum possible pixel value of the image, μ_y and $\mu_{\hat{y}}$ are the average of y and $\hat{y}$, respectively, σ_y^2 and $\sigma_{\hat{y}}^2$ are the variance of y and $\hat{y}$, respectively, $\sigma_{y,\hat{y}}$ is the covariance of y and $\hat{y}$, $c_1 = (k_1 R)^2$ and $c_2 = (k_2 R)^2$ are two variables to stabilize the division with weak denominator, R is the dynamic range of the pixel values, and $k_1 = 0.01$ and $k_2 = 0.03$ by default.

III. Experiment

Fig. 5 shows the experimental process of this study, which can be divided into four steps: data preparation, sample division, model training, and assessment.

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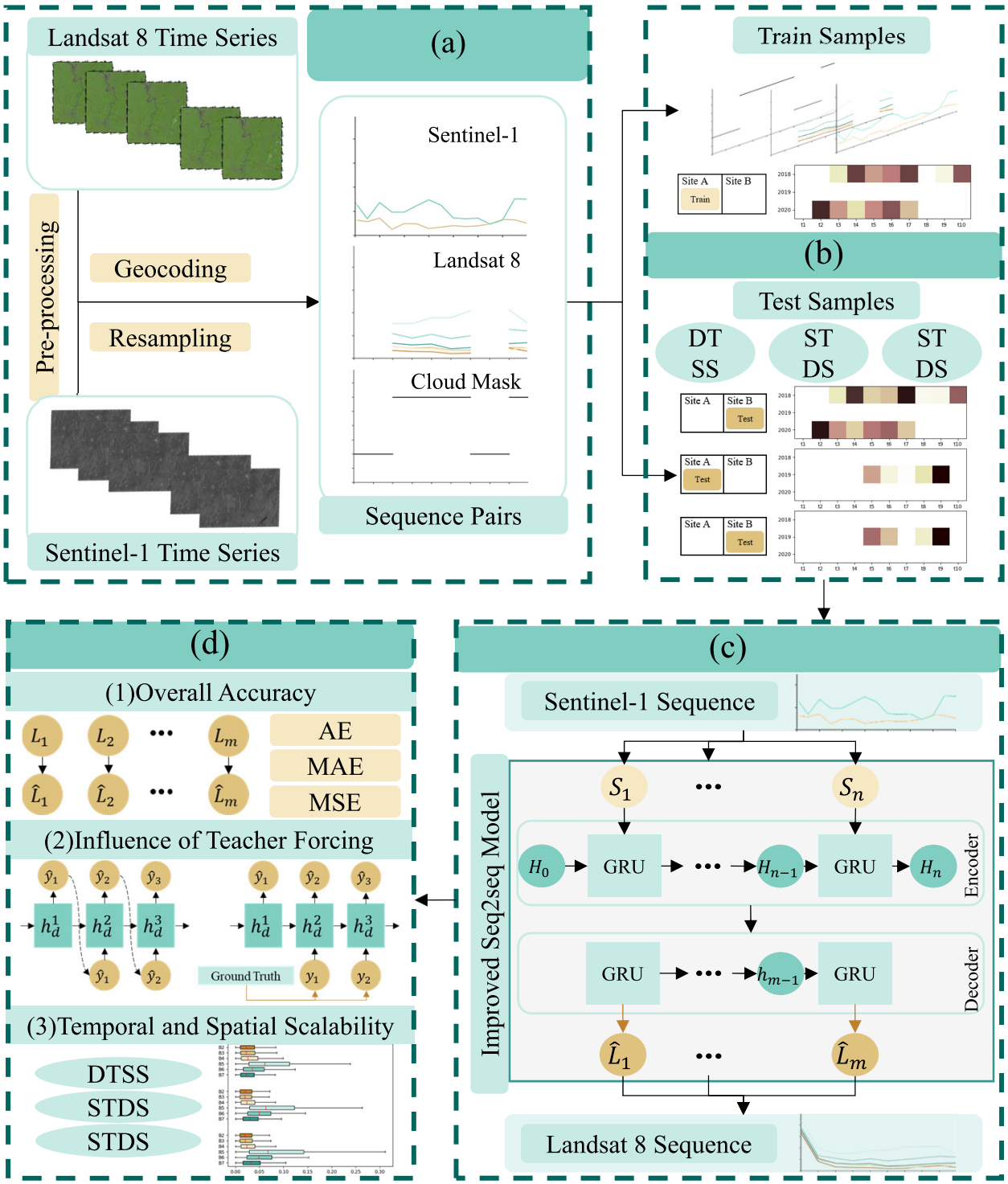


Fig. 5. Flowchart of the reconstruction of the optical time series based on the Seq2Seq model. (a) Data preparation. (b) Samples division. (c) Model training. (d) Assessment.

1) *Data Preparation* [see Fig. 5(a)]: After Landsat 8 and Sentinel-1 data were preprocessed, geocoding and resampling were performed to obtain pixel-to-pixel sequence pairs with the same geographic coordinate system and spatial resolution (30 m).
2) *Sample Division* [see Fig. 5(b)]: All pixel-to-pixel sequence pairs were divided into training and test samples by using the plot-level rice classification data. The test samples were further divided into three groups (see Section IV-B).
3) *Model Training* [see Fig. 5(c)]: We proposed an improved Seq2Seq model that integrated TF and the attention mechanism. The improved Seq2Seq model contained an encoder and a decoder, both of which were GRUs.
4) *Assessment* [see Fig. 5(d)]: We evaluated our model from three perspectives. First, the overall accuracy was evaluated by calculating the error between the outputs generated from Landsat 8 and cloudless Landsat 8 real-value sequences. Second, when using the trained

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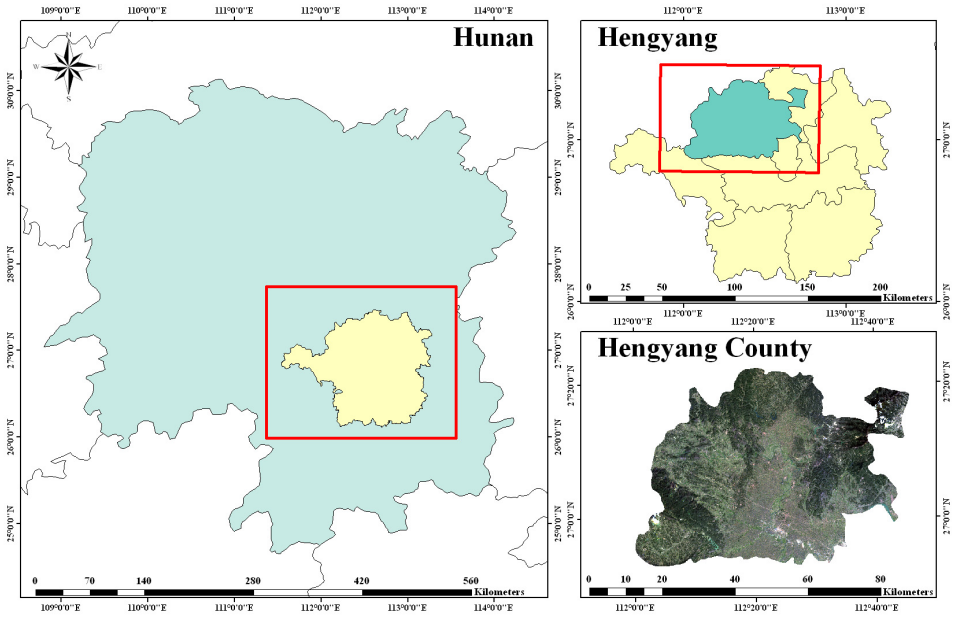


Fig. 6. Location of the study area in Hengyang City, Hunan Province, China.

Seq2Seq model, the error difference between with TF and without TF was compared to evaluate the effects of TF on the model. Third, three sets of test samples were used to evaluate the model performance in terms of the spatiotemporal scalability.

A. Study Area

Hengyang County (see Fig. 6) in the southcentral Hunan Province (110°52′ – 112°45′ E and 26°52′ – 27°23′ N) was selected as the study area. Hengyang County is characterized by a subtropical monsoon climate with abundant rainfall. Winters are warm and the spring temperature is high. The average annual temperature range is 17.5 °C – 18.1 °C and the average annual precipitation varies from 1200 to 1400 mm [72], [73]. Therefore, Hengyang County is a single-cropping rice (SCR) and double-cropping rice (DCR) mixed production area [74]. SCR is sown in early April, transplanted in late May/early June, and harvested in early September. DCR consists of early rice (sown in mid-March, transplanted from late April to mid-May, and harvested in late July or late August) and late rice (transplanted in July and harvested in October). According to the 2018 Hunan Provincial Bureau of Statistics (http://tjj.hunan.gov.cn/hntj/index.html), the seasonal precipitation distribution is given as follows: 27.49% (March–May), 30.10% (June–August), 28.26% (September–November), and 14.15% (December–January). The rice growth cycle is characterized by the frequent occurrence of clouds and rain, and thus, optical remote sensing images for this area are scarce.

B. Data and Preparation

1) Landsat 8 Optical Data: Landsat 8 carries an operational land imager (OLI) and a thermal infrared sensor (TIRS). The spatial resolution of Landsat 8 is 30 m and the temporal resolution is 16 days. We collected level-2 surface reflectance data from Landsat 8 OLI/TIRS corresponding to the rice-growing season in the study area from 2018 to 2020 from the United States Geological Survey (USGS) website (https://earthexplorer.usgs.gov/; 33 images, path/row 123/41, see Table SI).

The function of mask (Fmask) algorithm [76] was used to detect cloudy regions in Landsat 8 data and generate a cloud mask. First, Landsat top-of-atmosphere reflectance and brightness temperature data were used as inputs and rules based on the cloud's physical properties to separate potential cloud and clear-sky pixels. Next, the normalized temperature, spectral variability, and brightness probabilities were combined to produce separate probability masks for clouds over land and water. Subsequently, the potential cloud pixels and cloud probability mask were used to derive the potential cloud layer. By using the cloud mask, we removed invalid cloud regions and obtained cloud-free regions. After processing with the Fmask algorithm, the cloud mask sequence and cloudless Landsat 8 real-value sequence were obtained. It is important to note that the presence of outliers in the training data cannot be completely discarded, as the Fmask algorithm may contain errors. The performance impact of such errors will be discussed in Section VI.

2) Sentinel-1 SAR Data: Sentinel-1 is composed of a constellation of two satellites, Sentinel-1A and Sentinel-1B, which share the same orbital plane. Sentinel-1, with its C-SAR instrument, offers reliable, repeated, and wide-area monitoring. We obtained level-1 ground-range detected high-resolution (GRDH) images for the study area from 2018 to 2020 from the European Space Agency (ESA) website (https://scihub.copernicus.eu/dhus/#/home; 70 images, path/frames 11/81 and 11/86; see Table SI). All images were in the ascending pass. The GRDH product is obtained in interferometric wide mode with dual polarization (VH and VV, VH: transmitting vertically polarized waves and receiving horizontal polarized waves, VV: transmitting vertically polarized waves and receiving vertical polarized waves); it has a spatial resolution of 5 × 20 m and a revisit frequency of

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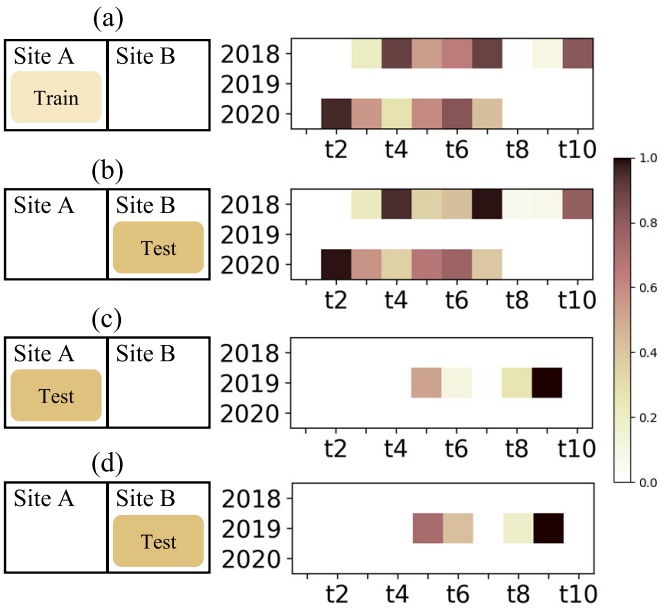


Fig. 7. Schematic of the temporal and spatial distributions of the training samples and three test samples. (a) Training samples. (b) STDS domain. (c) DTSS domain. (d) DTDS domains. The color scale represents the probability that all pixels have true values, 1.0 means that all pixels are cloud-free on that date, and 0.0 means that all pixels are cloudy on that date.

12 days. In this study, dual polarization was used to obtain the maximum amount of useful information from SAR data.

Because Sentinel-1 and Landsat 8 are two independent imaging systems, the spatial positions of images and spatial resolution differ. Therefore, it was necessary to perform geocoding and resampling. For Sentinel-1 data, we first performed a raster-to-point operation and then reprojected to the corresponding coordinate system of Landsat 8 for geographic registration. Sentinel-1 data were resampled using bilinear interpolation to reduce the spatial resolution to 30 m. The abovementioned geocoding and resampling operations were achieved by batch processing using ArcGIS software (https://www.esri.com/en-us/arcgis/).

3) Plot-Scale Rice Classification Data: Because of the intensive rice farming in Hengyang, the cultivated land is fragmented and general classification products cannot achieve the plot-scale rice classification. Therefore, this study first used the random forest classification algorithm to obtain coarse-classification rice vector data in Hengyang County. Subsequently, the plot-scale boundaries were vectorized on QGIS software in combination with visual interpretation based on coarse classification results and high-resolution images from Google Maps. A total of 47 216 plots were eventually acquired, involving 209 295 pixels. These vector data were used for the cropping of Landsat 8 and Sentinel-1 data, and training and test samples were obtained.

C. Sample Division

After cropping the cloudless Landsat 8 real-value and Sentinel-1 sequences using the aforementioned plot-level rice classification data, we obtained a total of 474 854 sequence pairs. We divided all sequences into training and test samples, yielding 365 759 pairs of training samples (77%) and

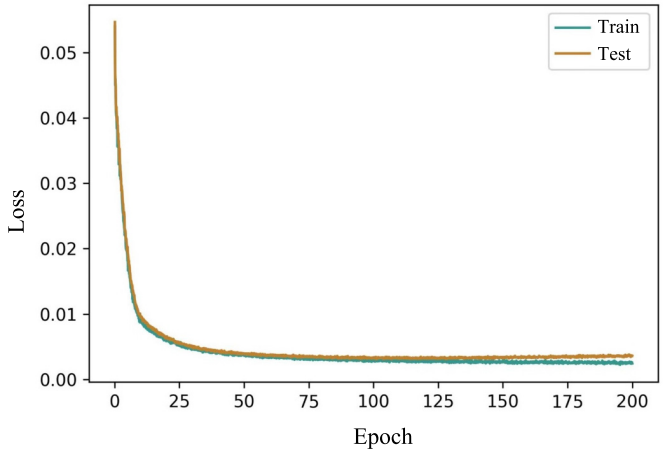


Fig. 8. Loss function curve of training and test samples.

109 095 pairs of test samples (23%). Note that we further divided the test samples into three test sets to evaluate the spatiotemporal scalability of the model: same time but different spatial domains (STDS), different times but the same spatial domain (DTSS), and both different times and spatial domains (DTDS). To further determine the correlation between the training samples and three test sets, we constructed a diagram displaying the spatiotemporal distributions (see Fig. 7). First, we randomly divided all available samples into two regions. We refer to these two regions as Sites A and B in Fig. 7. The training samples were from 2018 and 2020, from Site A. The STDS samples are also from 2018 and 2020 but originate from Site B. The DTSS test sample complements the training sample in time (2019) and space (Site A). The DTDS test sample did not coincide with the training sample in time and space, that is, it was obtained in 2019 at Site B. The STDS, DTSS, and DTDS test samples included 46 824, 53 334, and 8937 pairs, respectively.

D. Experimental Settings

All experiments were performed using an Intel Core i7-6700 2.6 GHz CPU, 64-GB memory, NVIDIA GeForce GTX 2080 GPU, and Pytorch framework in Python. Based on the model debugging results, the hyperparameters were set as follows: the optimizer was stochastic gradient descent (SGD), the learning rate was 0.005, and the hidden size was 2048. Finally, regarding the training and use of the model, we need to elaborate on the use of TFs and calculation of loss.

E. Model Training

With the model training method mentioned above (see Section II-B4), the model was trained in this study by random sampling, and the loss values were calculated for the validation samples simultaneously during the training process to investigate the training of the model (see Fig. 8).

At the beginning of the model training, the loss values of both the training and testing samples decreased rapidly as the model was trained: from the 1000th training to the 200 000th training, the loss values of the training and testing samples decreased from around 0.054 and 0.055 to around 0.0060 and

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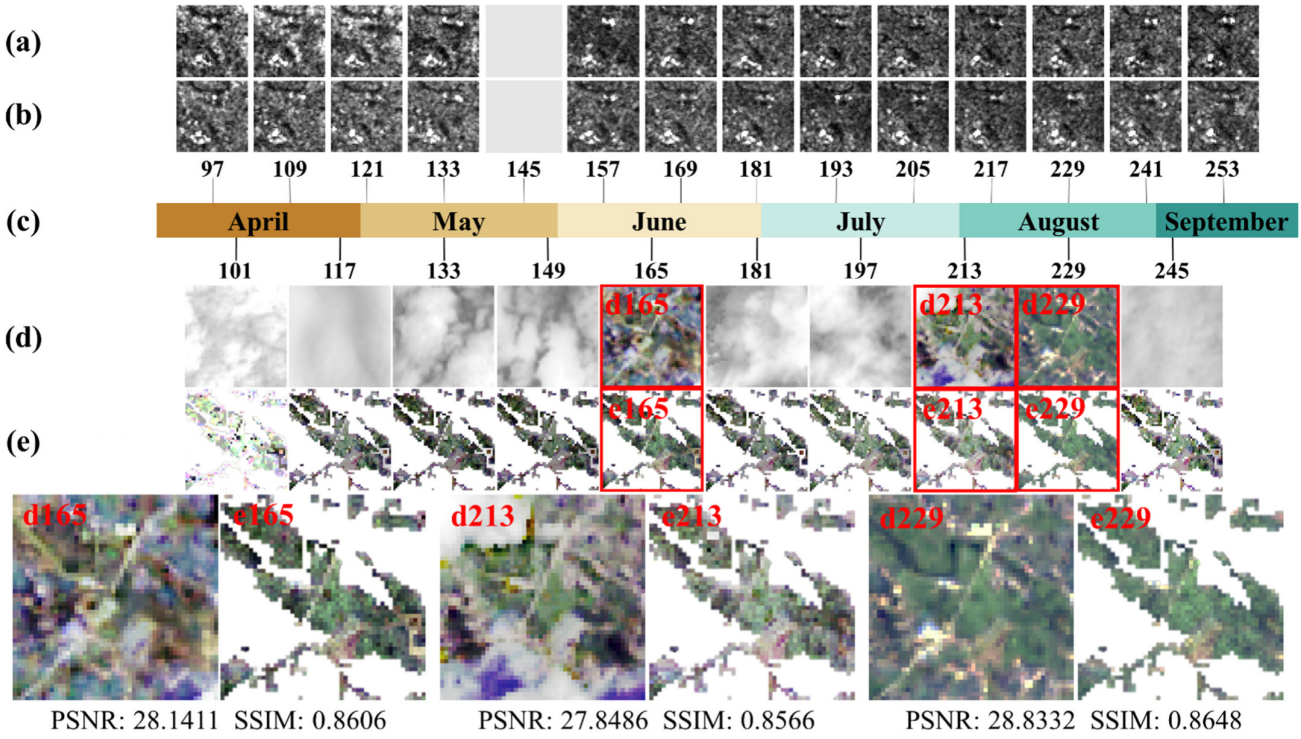


Fig. 9. Example for the reconstruction of optical time series based on the Seq2Seq model. (a). VH. (b) VV. (c) Date. (d) Landsat Real. (e) Landsat Generated.

0.0063, respectively. In the middle of the model training, the loss of the training and test samples decreases gradually as the model is trained. Finally, at the end of the training period, the loss values of the training and testing samples stabilized at around 0.0025 for the training samples and 0.0035 for the testing samples.

IV. Results

Fig. 9 shows the results for the study area as an example for reconstructing the optical time series based on the Seq2Seq model. Fig. 9(a) and (b) shows the temporal distribution of the used Sentinel-1 data (13 dates; Sentinel-1 data for the 145th day of 2019 could not be found on the ESA website), Fig. 9(c) shows the timeline and specific dates, and Fig. 9(d) shows the real Landsat 8 image, in which there are white clouds and various feature types, while Fig. 9(e) has only the pixels of cropland, which is after the mask of the rice classification data mask (ten dates). In Fig. 9, we also zoomed in on the images on days 165, 213, and 229 to visualize the reconstruction results. In addition, PSNR and SSIM were used as qualitative evaluation metrics to assess the reconstruction performance of the stitched images from the reconstructed pixel sequences. PSNR is above 27.8 and SSIM is above 0.85, indicating that the reconstructed images have good reconstruction effectiveness and structural similarity.

A. Overall Accuracy

In this study, a test set was used to test the trained model to quantitatively describe the cloud removal and imputation capabilities of the proposed model. We used the AE and MAE to represent the error distribution of the output-generated

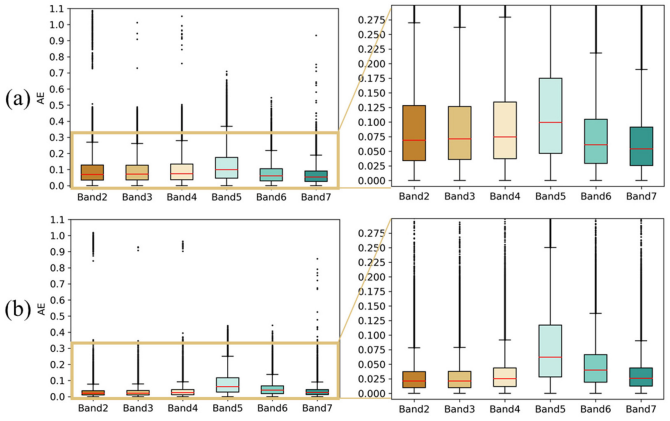


Fig. 10. AE of the six bands. (a) Original Seq2Seq. (b) Proposed method.

Landsat 8 sequence for each band. The box plot (see Fig. 10) shows the distribution of the absolute value of the difference between the real value (cloudless Landsat 8 real-value sequence) and generated value (output-generated Landsat 8 sequence) of each band and Table I lists the AE statistics for the different bands.

Overall, the proposed model yielded good results for bands 2, 3, 4, 6, and 7, but the accuracy of Band 5 was unsatisfactory (see Fig. 10). Approximately 72.7% of the generated Landsat 8 sequence values had an AE below 0.05. The MAE for all bands of the generated Landsat 8 sequence values was <0.0812. Bands 2, 3, 4, and 7 yielded good results in outputting the generated Landsat 8 sequence, with an AE below 0.05 ($>80.52\%$). Band 6 was determined to be the least effective. However, the generated Landsat 8 sequence values of Band 6 with an AE below 0.05 also account for more than

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TABLE I
MAE VALUES AND AREA PERCENTAGE OF EACH AE CLASSIFICATION IN SIX DIFFERENT BANDS

	AE	Mean	0-0.025	0.025-0.05	0.05-0.1	0.1-0.15	>0.15
Band 2	Original Seq2Seq	0.0946	9.46%	17.65%	20.08%	27.53%	15.01%
	Proposed method	0.028	57.00%	29.26%	11.72%	1.64%	0.38%
Band 3	Original Seq2Seq	0.0929	9.29%	17.51%	17.01%	31.01%	15.48%
	Proposed method	0.0276	56.95%	28.96%	11.94%	1.77%	0.38%
Band 4	Original Seq2Seq	0.0971	9.71%	16.67%	17.50%	28.73%	16.19%
	Proposed method	0.0318	49.89%	30.63%	16.55%	2.36%	0.56%
Band 5	Original Seq2Seq	0.1205	12.05%	13.65%	13.30%	23.32%	17.76%
	Proposed method	0.0812	22.10%	19.99%	26.81%	14.75%	16.36%
Band 6	Original Seq2Seq	0.0741	7.41%	21.40%	20.17%	31.28%	16.27%
	Proposed method	0.0465	32.26%	28.07%	32.09%	6.46%	1.12%
Band 7	Original Seq2Seq	0.0635	6.35%	24.58%	22.01%	32.90%	14.82%
	Proposed method	0.0311	48.75%	32.55%	16.78%	1.63%	0.29%

60.33% and those with an AE below 0.1 account for more than 92.42%. Band 5 showed the worst performance, with the largest MAE (0.0812) and largest percentage (16.36%) of pixels with an AE above 0.15. Due to the effect of the cloud detection method, a small fraction of anomalous error is apparent (see details in Section V).

The real and generated values of each time range are shown in Fig. 11. The generated-value sequences fit the real-value sequences well, indicating that the proposed model had a good generation ability. The yellow range in Fig. 11(a) represented outliers in the real Landsat 8 sequences after cloud removal. Abnormal values accounted for less than 5% of all cloudless Landsat 8 real-value sequences. The light green range in Fig. 11(a) was the range of all sequence values at the corresponding time. Fig. 11(b) shows the reflectivity range of the six bands of the output-generated Landsat 8 sequence. The output-generated values of each band fall within a narrow range. Similar to the results described in Fig. 10, Band 5 had a wider range of generated values than that of the other bands, indicating that several values were generated, which deviate from the reference line, and that the proposed model was less effective in Band 5.

Compared with the original Seq2Seq method, the proposed method in this study improves the reconstruction effect for each band, especially for bands 2, 3, and 4 (MAE decreases by 0.0666, 0.0653, and 0.0653, respectively). The green–blue margin in Fig. 11(b) is larger, which means that the fluctuation of the reconstructed data based on the original method is more serious, and a large amount of data seriously deviates from the real data. In contrast, the green–blue margin in Fig. 11(c) is small, i.e., all the generated values are close to the median, indicating that the results of the proposed method are more stable, and the majority of generated values are close to the real data, while only a small amount of data deviates from the real data.

B. Effect of TF

In this study, the TF method was introduced to the use stage of the model to improve the accuracy of the output-generated Landsat 8 sequence. We conducted two sets of experiments for comparison: 1) TF was used and 2) TF was not used. The results (Table II and Fig. 12) obtained by using TF are generally better than those derived without TF and the MSE value is lower. The accuracy improvement was the most notable in Band 5, with an increase of 9.46%, followed by bands 3 and 6 (9.33% and 9.21%, respectively). Band 7 showed the worst effect with respect to the improvement of the accuracy, with an increase of only 5.97%.

To visualize the effect of the use of TF on the accuracy, we randomly selected a pixel and visualized the six bands of its time series (see Fig. 13). By comparing the two resulting sequences, the shapes of the two sequences were found to be relatively similar. By comparing the resulting sequences generated by the two methods in all bands with the real values at $t7$, the results generated with TF are closer to the real values than those generated without TF in all bands, with Bands 4 and 5 showing more pronounced effects.

Consider the case in which the generated value at the fourth moment was higher than the real value: as shown in the highlighted area of the yellow band in Fig. 13(d), when the TF method is not used, the results at $t5$ and $t6$ show an upward trend, whereas when the TF is used, the results at $t5$ and $t6$ are significantly lower than the results without the TF.

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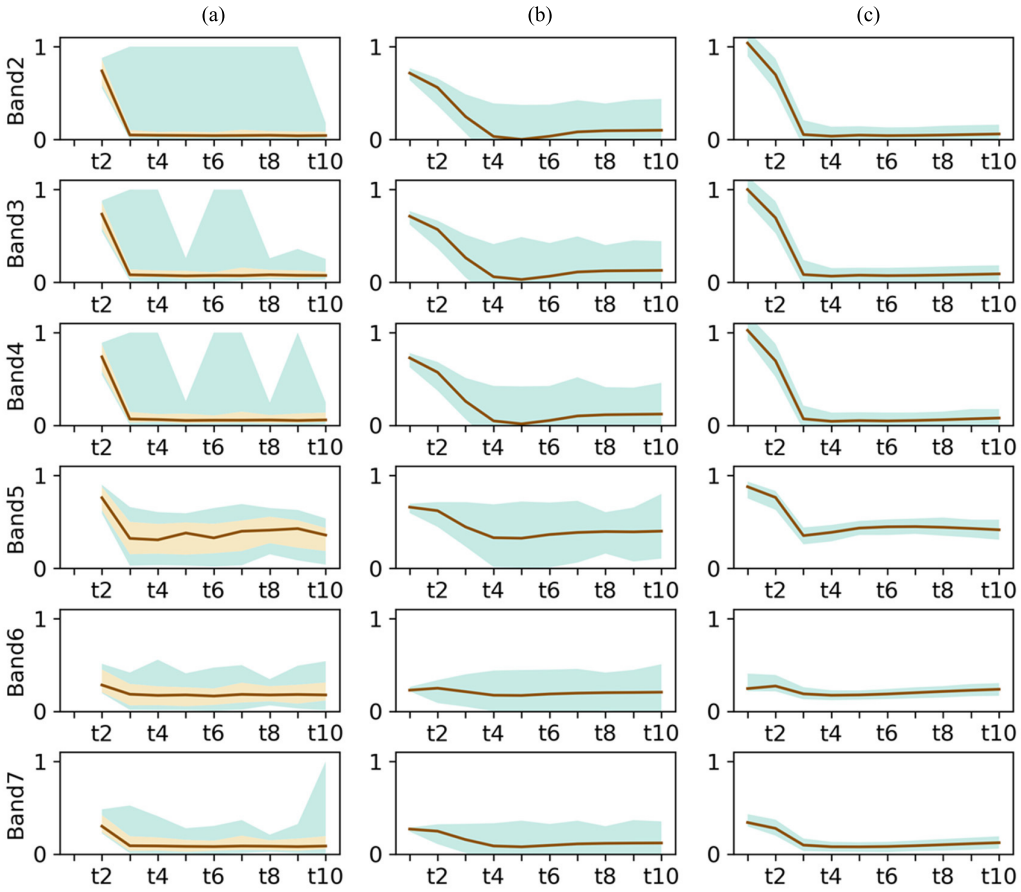


Fig. 11. Comparison between real and generated values for six bands based on the proposed Seq2Seq model. The brown line is the median of each corresponding time. The yellow margin is the range of sequence values excluding outliers at the corresponding time, and the green–blue margin is the range of all sequence values at the corresponding time. (a) Real value. (b) Original Seq2Seq. (c) Proposed method.

TABLE II
COMPARISON OF THE MSES OF DIFFERENT BANDS WITH AND WITHOUT TF

MSE	Band 2	Band 3	Band 4	Band 5	Band 6	Band 7
No use TF	0.00234	0.001673	0.002054	0.010777	0.003576	0.001694
Use TF	0.002158	0.001517	0.001916	0.009758	0.003247	0.001592
Improvement	7.76%	9.33%	6.73%	9.46%	9.21%	5.97%

This suggests that the use of the TF method better constrains the generation of results at the next moment, and the model may capture anomalous information based on the use of TF.

C. Temporal and Spatial Scalability

To ascertain the robustness of the proposed model in the temporal and spatial domains, we divided the test samples into three categories according to the differences of the training samples: STDS, DTSS, and DTDS. Based on Table III, for STDS, Band 7 has the highest accuracy (0.00156) and Band 5 has the lowest accuracy (0.0106). For DTSS, Band 3 has the highest accuracy (0.00097) and Band 5 had the lowest accuracy (0.0117). For DTDS, the accuracy of Band 3 was the highest (0.00106) and the accuracy of Band 5 was the lowest (0.0142), similar to DTSS. Overall, the accuracies of bands 2, 3, 4, and 7 of all three test samples were high, followed by Band 6 (see Fig. 14). The accuracy of Band 5 was the lowest, but the overall accuracy of the three test samples was within a reasonable range. A sharp drop in the accuracy of a certain test sample set was not observed. This demonstrated that the proposed model was robust in terms of spatiotemporal scalability.

V. DISCUSSION

During the cloud removal based on the Seq2Seq model, different hyperparameter combinations exhibited different performances. Hyperparameters can generally be fixed prior to model training. They can be divided into two main categories. One category refers to the type of optimizer, the number of epochs, and learning rate. Such hyperparameters are generally

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TABLE III
COMPARISON OF THE MSES OF DIFFERENT BANDS FOR STDS, DTSS, AND DTDS

MSE	Band 2	Band 3	Band 4	Band 5	Band 6	Band 7
STDS	0.002144	0.001706	0.002159	0.010613	0.003032	0.001562
DTSS	0.001921	0.00097	0.001346	0.011703	0.003819	0.001669
DTDS	0.001329	0.001056	0.001327	0.014182	0.003987	0.001937

∗different spatial domain (STDS), different time but the same spatial domain (DTSS), and both different time and spatial domains (DTDS)

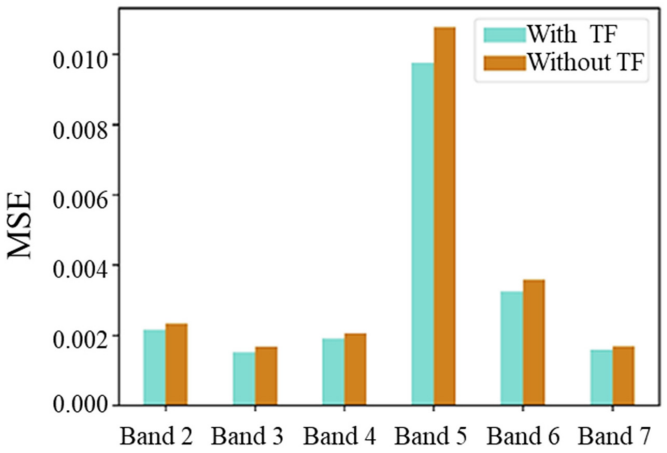


Fig. 12. Comparison of the MSEs of the output-generated Landsat 8 sequence with and without TF.

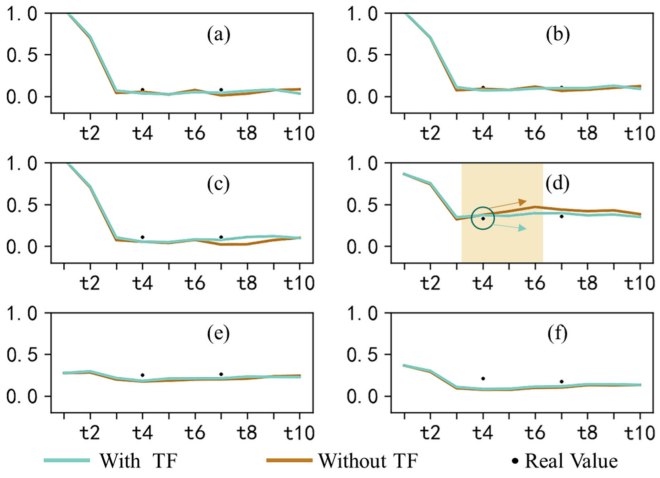


Fig. 13. Comparison of the output-generated Landsat 8 sequence with and without TF based on the use of a random pixel as an example. (a) Band 2. (b) Band 3. (c) Band 4. (d) Band 5. (e) Band 6. (f) Band 7.

obtained based on experience. The other category consists of hyperparameters in the model framework that must be adjusted based on experiments such as the hidden size. In the Seq2Seq model, the number of hidden layer nodes determines the vector that can be used to record sequence information and the size of the context vector, which directly affects the ability of the proposed model to extract and save time series information. A larger number of hidden layer nodes enable the model to record more information obtained from Sentinel-1, but it also significantly increases the training time of the model.

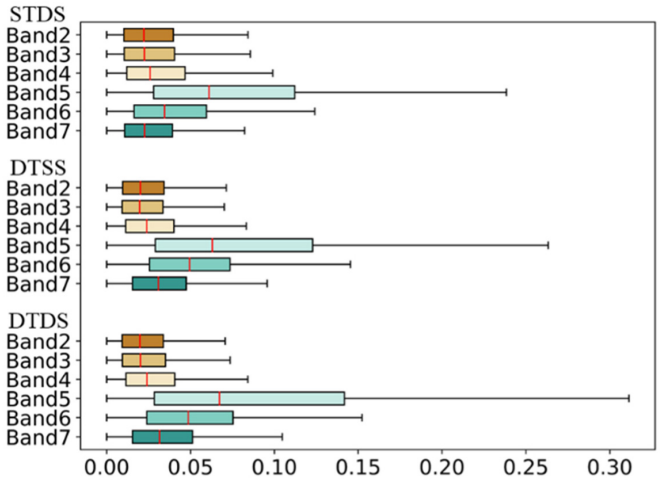


Fig. 14. Comparison of the AE of output-generated Landsat 8 sequence data for STDS domain, DTSS domain, and both DTDS domains.

It is worth exploring the tradeoff between longer sequence information records and shorter training time to determine the optimal solution.

The common deep learning methods used for data reconstruction are CNN [42], ResNet [40], and GAN [34], [35], which all reconstruct monotemporal images with the help of the image–image reconstruction approach. Recently, several scholars have performed time series–time series reconstruction; however, their studies focused on the equal-length vegetation index (such as NDVI) reconstruction. The so-called equal length is an artificially regularized sequence of unequal lengths obtained due to different reentry cycles in the same period to the same length by tedious preprocessing. For example, previously, a common temporal grid was defined for optical and SAR datasets [57]. The target grid was the ascending or descending SAR time series with the lowest average time lag (in days) for optical acquisitions. The optical and remaining SAR data were then resampled to this temporal grid using the nearest neighbor technique. The proposed method can deal with not only equal-length sequences but also unequal length sequences. In addition, the output is a full-band time series. Therefore, the proposed method has the advantages of flexibility, easy sample acquisition, and spatiotemporal scalability compared to the image-to-image approach.

Although many methods are available, accurate cloud and shadow detection remain a challenge in optical remote sensing [9]. In this study, a cloud mask was used for the loss calculation. However, the cloud removal algorithm (Fmask)

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used in this study did not completely remove the clouds to form an accurate cloud mask. As described in Section IV, outliers were detected (less than 5% of the total pixels). However, these outliers did not cause significant problems with respect to the results. Fig. 11 shows that the yellow areas in the left picture, which represent outliers, cannot be observed in the results on the right. This shows that our model overcomes these outliers. However, inaccurate cloud masks affect the loss calculation, which is difficult to estimate. In future research, the focus should be placed on more effective cloud removal algorithms to obtain accurate cloud masks and reduce potential negative effects.

However, this study considers rice, which is only a reconstruction of a single land cover type and does not explore the performance of the model under multiple land cover types. There are cases of abandonment and replacement of rice, and there may be a small number of other land cover types mixed in, resulting in reduced reconstruction accuracy. In addition, the subtle changes in the vegetation itself can reflect changes in Band 5 of Landsat 8, but they may not be effectively reflected in the SAR data. Compared with SAR images, Band 6 in Landsat is more sensitive to the vegetation water content. Therefore, it is unsatisfactory to accurately model the relationship between Band 5 or Band 6 in Landsat 8 and the SAR data. To solve this problem, it could be feasible to proceed by comparing the differences between the spectral curves of vegetation under different growth cycles and different biological conditions while reconstructing based on the multisource remote sensing images, i.e., SAR and optical images simultaneously as auxiliary data sources. The above problems should be addressed with efforts in future work.

VI. Conclusion

The main aim of this study was to improve the Seq2Seq model to reconstruct the optical image sequence based on SAR time series information and reduce the number of samples based on the modification of the loss calculation method. The major findings are given as follows.

1) The improved Seq2Seq model can be used for the cloud removal and reconstruction of optical full-band time series using unequal length SAR time. Approximately 72.7% of the generated Landsat 8 sequence values have an AE of less than 0.05. The reconstruction effect of each band was given as follows: Bands 2, 3, 4, and 7 were the best, followed by Band 6, and Band 5 was the worst.
2) The introduction of the TF-generated-value sequence in the Seq2Seq model use stage will be closer to the real value and the accuracy of each band improves to different degrees. This shows that the generated-value sequence can be better constrained by introducing TF in the model use stage.
3) The MSE of the three test samples, that is, STDS, DTSS, and DTDS, is similar. All errors are lower than 0.015, which shows that the proposed model has good robustness in both the time and space dimensions.

These findings provide a basis for the use of time series pairs with unequal lengths in cloud removal and vegetation index reconstruction. Although we only reconstructed the Landsat 8 time series from Sentinel-1, the reconstruction of other SAR time series to optical series is theoretically feasible. Our results show that the reconstruction of SAR time series to optical series can be used to construct time series data with high temporal and spectral resolution. In future research, different hyperparameter combinations should be considered to improve the reconstruction accuracy.

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Tao Peng received the B.S. degree in geographic information science (GIS) from the China University of Geosciences, Beijing, China, in 2018, where he is currently pursuing the M.S. degree in surveying science and technology.

His research interests include deep learning and time series reconstruction.



Meiling Liu received the B.S. degree from Hunan Normal University, Changsha, China, in 2002, and the M.S. and Ph.D. degrees from the China University of Geosciences, Beijing, China, in 2006 and 2011, respectively.

She is currently a Professor with the School of Information Engineering, China University of Geoscience. Her research interests include developing models and methods for studies of natural resources, ecological, and environmental systems by remote sensing and Geographic Information Science (GIS).



Xiangnan Liu received the B.S. degree in geography from Hunan Normal University, Changsha, China, in 1987, and the M.S. and Ph.D. degrees in remote sensing and geographic information systems from Northeast Normal University, Changchun, China, in 1990 and 1996, respectively.

In 1990, he joined Northeast Normal University as a Faculty Member. Since 2006, he has been a Professor with the School of Information Engineering, China University of Geosciences, Beijing, China. He has authored or coauthored seven books and more than 100 articles. His research interests include modeling, quantitative analysis of natural resources, ecological, and environmental systems by remote sensing, Geographic Information Science (GIS), and geostatistics.



Qian Zhang received the B.S. degree in geographic information science and the M.S. degree in survey engineering from the China University of Geosciences, Beijing, China, in 2018 and 2021, respectively, where she is currently pursuing the Ph.D. degree in surveying science and technology.

Her research interests include synthetic aperture radar (SAR)-to-optical translation and deep learning.



Ling Wu received the B.S. degree in geographic information systems and the M.S. and Ph.D. degrees in cartography and geographic information engineering from the China University of Geosciences, Beijing, China, in 2007, 2010, and 2013, respectively.

He held a post-doctoral position with the Institute of Remote Sensing and Geographic Information Science (GIS), Peking University, Beijing, from 2013 to 2016. He is currently an Associate Professor with the School of Information Engineering, China University of Geosciences. His research interests include time series analysis and scale transformation of remote sensing products.



Xinyu Zou received the B.S. degree in geographic information systems and the M.S. degree in survey engineering from the China University of Geosciences, Beijing, China, in 2014 and 2018, respectively, where he is currently pursuing the Ph.D. degree in surveying science and technology.

His research interests include time series analysis, vegetation phenology, and spatiotemporal fusion.

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